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1.0 Introduction

Cardiovascular diseases are one of the most significant health challenges worldwide, contributing to millions of deaths each year. Early detection and accurate prediction of cardiovascular risk in an individual can save that individual from costs of treatment, pain, and stress, leading to an overall healthier and happier life. However, while there are ways to predict a person's cardiovascular risk - low, medium, high - it is difficult to do so accurately risk due to the wide array of complex and multifaceted factors - demographic, lifestyle habits, and pre-existing conditions - involved and the dissimilarities amongst us. With the rise of machine learning and AI, diagnosis of cardiovascular risk has been made easier and smoother, albeit the predictions still require professionals' insights. Nonetheless, machine learning offers a powerful set of tools capable of analysing large datasets and uncover patterns that may not be evident under statistical methods; machine learning can help doctors and researchers to learn and uncover unsuspected correlations and new trends, allowing them to make informed decisions.

To solve this issue, machine learning techniques are used to analyse a dataset that includes features such as gender, age, family history, and more. By employing multiple models, we can determine which model does the best in predicting cardiovascular risks in individuals under specific conditions, showing us the most significant factor while uncovering patterns and previously unknown correlations within the data. This process involves selecting and training models followed by fine tuning them. The ultimate goal is to narrow down the models that is both robust and reliable in assisting healthcare professionals and researchers in identifying at-risk individuals, thereby facilitating timely medical interventions.

The objectives of the project are to find out the relevant factors that contribute to cardiovascular risk of an individual, develop and optimise machine learning models that can

accurately predict the cardiovascular risk level of an individual. This is achieved through data exploration and pre-processing, model selection and training, model tuning and evaluation, and result analysis and interpretation. In data exploration and pre-processing, the dataset is visualised and processed to handle missing values, outliers, and noise, to ensure only cleaned relevant data is used for model training. Furthermore, in model selection and training, various models are chosen and trained to determine the most effective mode. The selected models are then fine-tuned and analysed once more.

2.0 Methods

2.1 Dataset description

The dataset “dataset.csv” contains information about the characteristics and lifestyles of individuals to analyse individuals cardiovascular risk based on how they live and their background. The dataset contains **2100** rows, **17** attributes and **1** label.

No	Feature	Unit	DataType	Values
1	Gender	-	String	Female, Male
2	Age	-	Integer	17
3	Height	cm	Float	172.2
4	Weight	kg	Float	70
5	Family History	-	String	yes, no
6	Alcohol	-	String	none, low, medium, high
7	Junk Food	-	String	yes, no
8	Vege Day	-	Int	1, 2, 3
9	Meals Day	-	Int	1, 2, 3, 4
10	Snack	-	String	sometimes, frequently, always, no
11	Smoking	-	String	yes, no
12	Water Intake	L	Int	2.72
13	Transportation	-	String	walk, bicycle, motorcycle, car, bus
14	Exercise	-	Int	0, 1, 2, 3
15	TV	-	String	rare, moderate, often
16	Income	-	Int	14046
17	Discipline	-	String	yes, no

These features encapsulate various aspects of an individual, such as lifestyle, health metrics, and demographic profile. The label is categorised into *low*, *medium*, and *high*. However, there are no column that acts as the unique identifier.

There are **6 features** with **nominal data**, **3 features** with **ordinal data**, **5 integral features**, and **3 features** with **floating point values**.

2.2 Data exploration and visualization

Upon exploration, it is found that there are no null values in the dataset. Since there are 17 features. Before moving on to analysing and visualizing the dataset, it is important to extract relevant features first. We have decided to only include **gender, age, height(cm), weight(kg), family_history, alcohol, junk_food, snack, smoking, transportation, exercise, tv, and discipline**. Height and weight will be combined to become BMI as, individually, doesn't give much information.

We chose these features because they have more **direct** relationships with cardiovascular risks compared to `vege_day`, `meals_day`, `water_intake(L)`, and `income`. The meaning of the values in `vege_day` is ambiguous and do not synchronise with the values in `meals_day`. The volume of water intake does influence the risk of cardiovascular disease but, its relationship with cardiovascular risk is often indirect and influenced by other factors. We chose to exclude `income` as we have features such as `junk_food`, `smoking`, `exercise`, and `TV` that can serve as proxies for socio-economic status. We have also decided to focus on medical factors and lifestyle factors with more direct relationships.

2.2.1 Univariate Analysis

Cardiovascular Risk

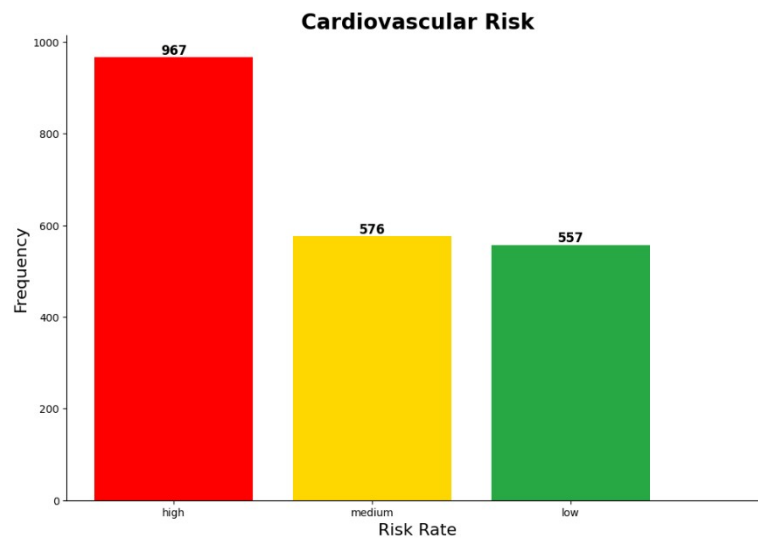


Figure 2.1: Cardiovascular Risk Bar graph

Most individuals, 967, in the dataset have high risk of developing cardiovascular diseases. The distribution of individuals with low and medium cardiovascular risk is rather evenly distributed with minor differences.

Numerical Data

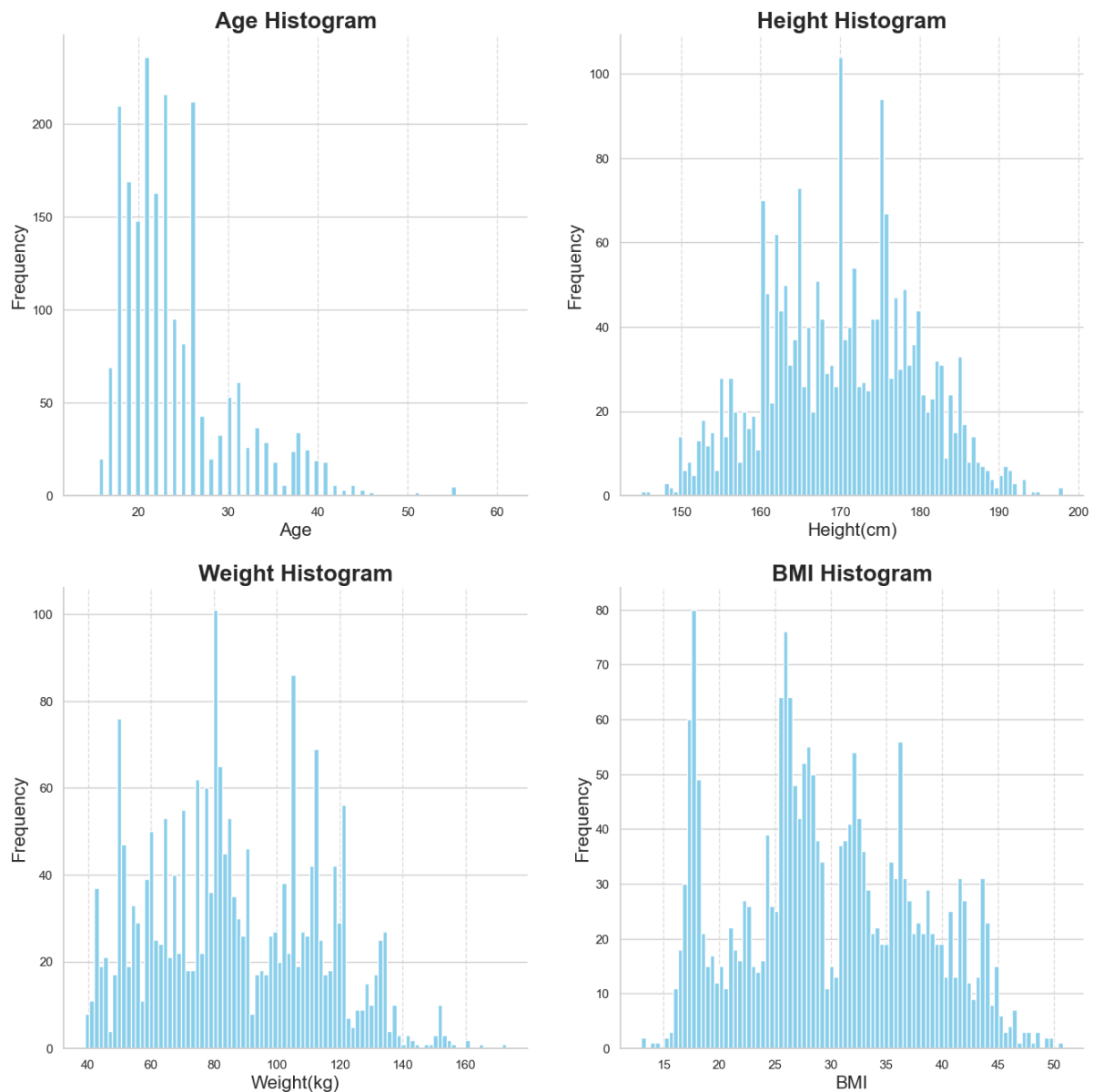


Figure 2.2: Numerical Data Histogram

It is found that the average age in the dataset is 24 years old with the youngest being 14 and the oldest being 61 years old. The distribution of age is right skewed, suggesting that the demographic of the dataset is focused on teenagers and working adults.

The average height is 170.15cm as shown in the histogram with the shortest being 145cm and the tallest being 198cm. The histogram shows that the distribution of height is a rather normal distribution with some spikes.

The average weight is found to be 86.57kg with the lightest at 39kg and the heaviest at 173kg. The distribution of weight is slightly right skewed with some seemingly extreme weights.

The histogram denotes a multimodal distribution, as shown by the presence of multiple peaks. There are peaks between 15 and 20, between 25 and 30, and between 30 and 40. This suggests that the dataset may represent several underlying subpopulations or distinct clusters with different BMI ranges.

Categorical Data

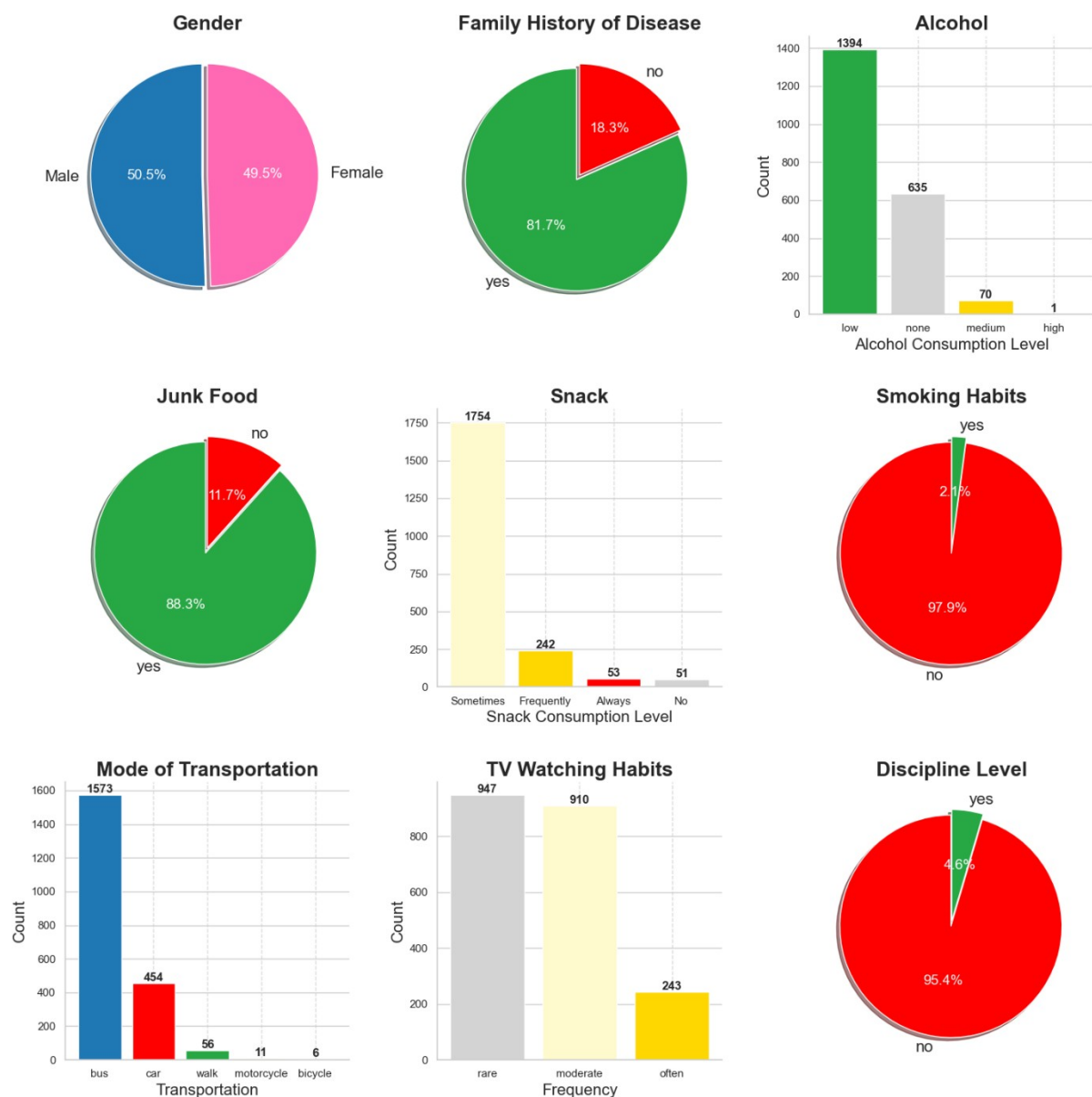


Figure 2.3: Categorical data plot matrix

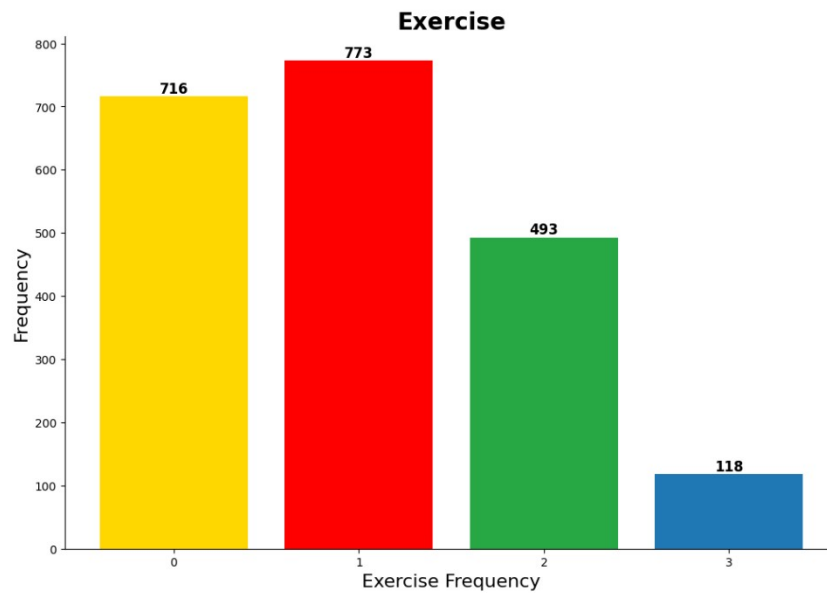


Figure 2.4: Exercise Bar chart

The data distribution for the gender feature is evenly distributed with 50.5% male and 49.5% female. This suggests that both genders are evenly represented in the dataset. This can minimise potential bias in the machine learning model, preventing the model from skewing towards one gender over the other. Majority of the individuals, 81.7%, have someone in their family with cardiovascular disease, whereas only 18.3% individuals without. The alcohol feature has its “low” value dominate the rest by a wide margin. This may introduce bias to the model when determining the cardiovascular risk of an individual. 88.3% individuals admitted to indulging in junk food while only 11.7% of individuals said they do not.

The number of individuals who claimed that they have snacks sometimes are much larger than the number of individuals who claimed otherwise by a large margin. Similar to the alcohol feature, due to the lack of even representation, this may cause the model to skew towards individuals who sometimes have snacks. Another feature with uneven distribution, smoking, records 97.9% no to smoking of and only 2.1% yes.

The most popular form of transportation in the dataset is a bus, a sedentary transportation, with 1573 individuals using it. In contrast, only 6 individuals ride the bicycle.

This may suggest that the data was gathered from urban areas where buses are more common, and cycling may be less prevalent. 947 individuals rarely watch TV, 910 watch TV moderately, and only 243 watch TV frequently. Finally, most individuals, 95.4%, claimed they are not discipline while only 4.6% claimed they are. This is supported by the exercise feature data distribution; 716 individuals do not exercise, 773 individuals rarely exercise, 493 moderate exercises, and 118 regularly exercises.

2.2.2 Bivariate Analysis

Age and Cardiovascular Risk

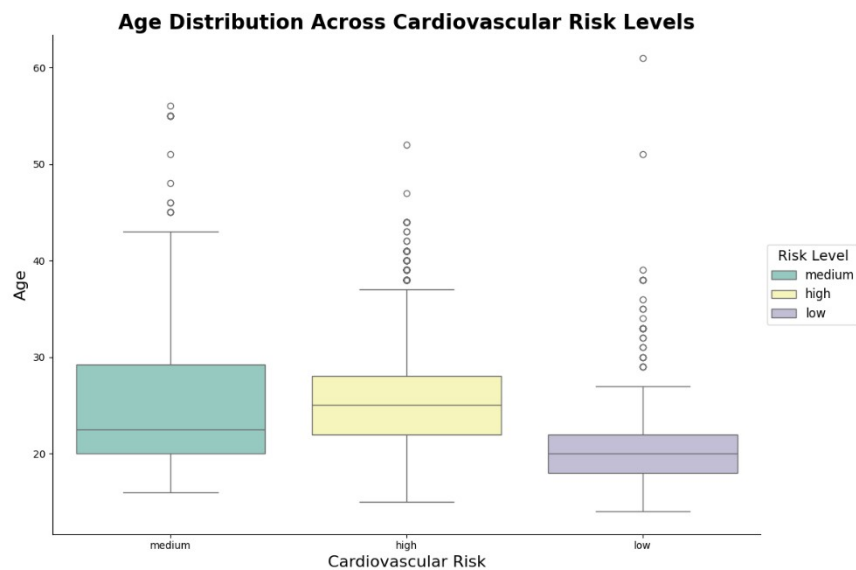


Figure 2.5: Age vs Cardiovascular risk boxplot

The boxplot for the low-risk category shows that most individuals in this group are younger, with the median age being lower than the other categories. The range of ages in this group is also narrow, with a few outliers extending to older ages.

On the other hand, the medium-risk category displays a wider age range, with its median age positioned slightly higher than the low-risk boxplot. The distribution of age also suggests that individuals in this group are more diverse, with several outliers.

Finally, the high-risk group shows an age distribution similar to the medium-risk group, albeit slightly skewed towards older individuals. The median age in this group is also higher, indicating that older individuals are more likely to fall into the high-risk category. The presence of outliers at younger ages suggests that some younger individuals also face significant cardiovascular risk.

BMI vs Cardiovascular Risk

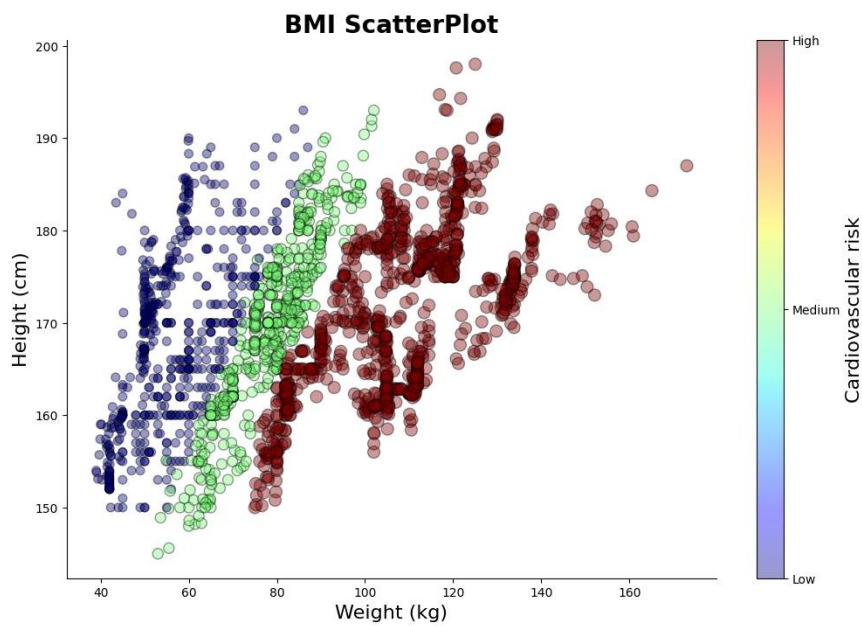


Figure 2.6: BMI vs Cardiovascular Risk scatter plot

The scatter graph suggests correlations between BMI and cardiovascular risk. Based on the scatter graph, individuals with lower BMI have lower risks of developing cardiovascular diseases while individuals with higher BMI have higher cardiovascular risk.

Gender vs Cardiovascular Risk

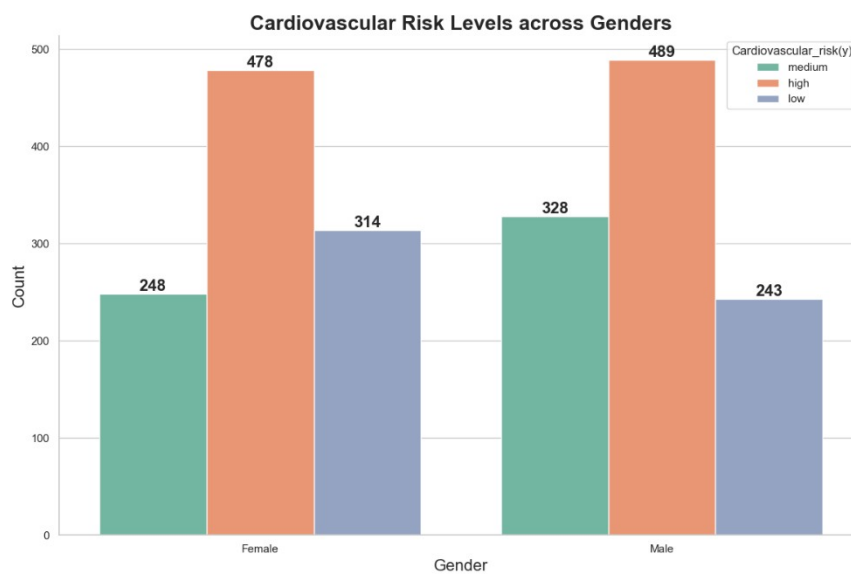


Figure 2.7: Cardiovascular Risk across Genders

Both females and males have the highest count in the high-risk category, with counts reaching 500, indicating that high cardiovascular risk is prevalent in both genders. However,

there are more medium-risk category counts in males compared to females. There are also lesser males with low-risk category compared to females.

Lifestyle vs Cardiovascular Risk

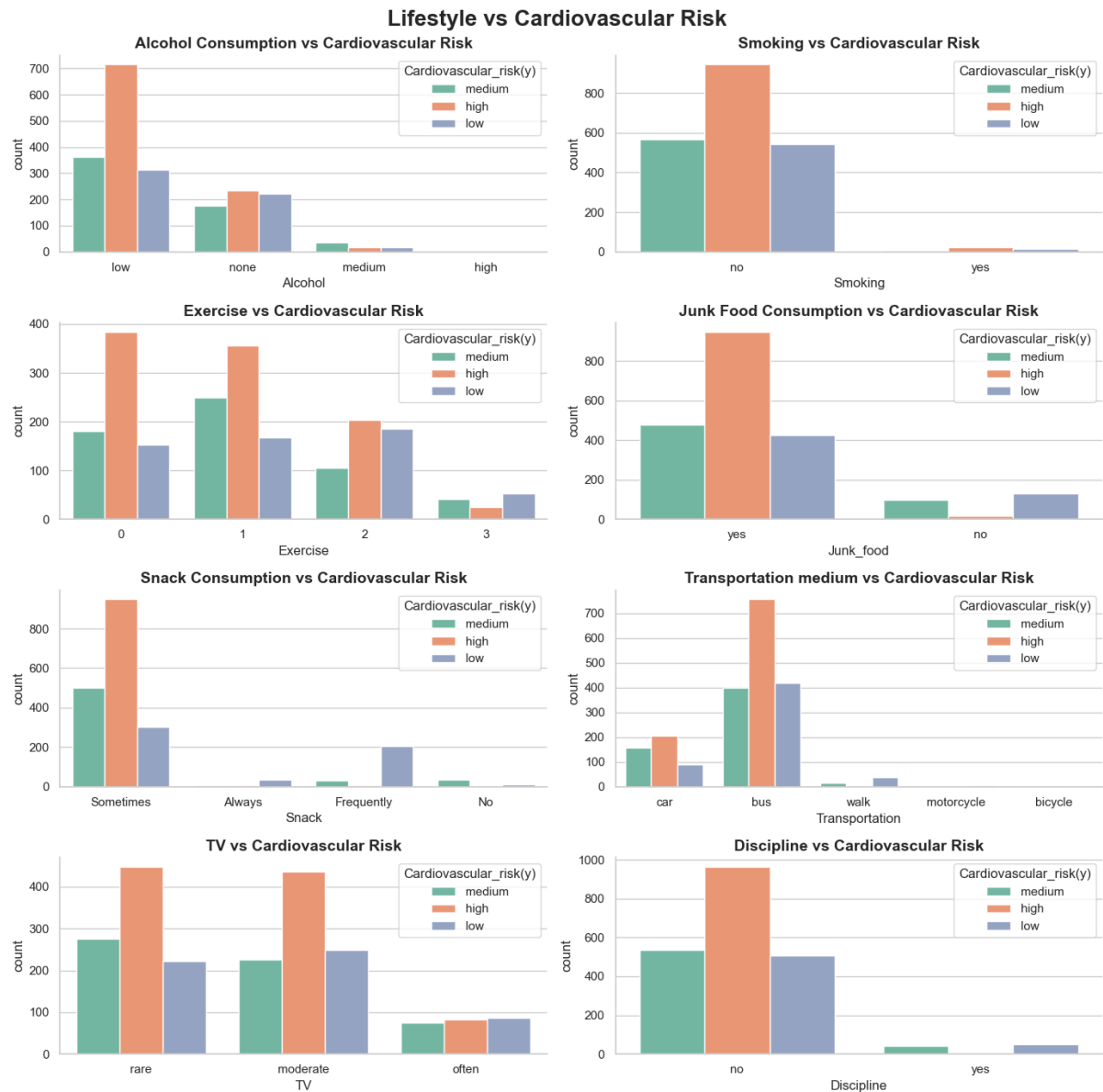


Figure 2.8: Lifestyle vs Cardiovascular Risk

From the group plot above, it can be observed that a significant portion of individuals who consume low volumes of alcohol have higher cardiovascular risks. This result contradicts the research done that show a link between alcohol consumption rates and cardiovascular risks. Nonetheless, this trend may be attributed to the higher number of individuals who consume low amounts of alcohol in the dataset rather than the direct effects of alcohol consumption on

cardiovascular risks. This applies to smoking, snack consumption rate, and tv as well; majority of individuals who do not smoke, indulge in snacks sometimes, and rarely watch tv fall in the high cardiovascular risk category despite having multiple research that say otherwise.

Additionally, more individuals who do not exercise have fall into high cardiovascular risk categories in contrast to individuals who regularly exercise. Individuals who take passive transportation have much larger cardiovascular risk as opposed to those who take active transportation. Majority of individuals who admitted to consuming junk food fall into high cardiovascular risk category. Finally, individuals believe they do not have discipline have higher cardiovascular risk compared to those who do.

2.2.4 Potential Issues

The analysis reveals several patterns between lifestyle factors and cardiovascular risk, such as alcohol consumption, and cardiovascular risk levels. For instance, a significant proportion of individuals who consume low amounts of alcohol are categorized as having a high cardiovascular risk. However, it can be conjectured that these observed relationships may be driven by the distribution of the data rather than representing genuine causal links between the features and the label.

This highlights the importance of careful consideration when interpreting correlations in data, as they might be influenced by the distribution rather than a direct cause-and-effect relationship. To fix this issue, more data must be gathered to ensure that each group has equal representation and the relationship between the feature and label is more realistic and accurate.

2.3 Data Preprocessing

Due to the highly skewed distribution in numerical data (age, height, weight, and exercise), there are a lot of outliers in the dataset. Before removing outliers, height and weight are first combined into BMI as BMI gives a more meaningful measurement of body fat relative to height and weight. If outliers from weight and height are removed before combining into BMI, we may overlook how these two features interact. If observed separately, the values in weight and height may seem extreme but, combined, may look normal.

Then, the dataset is divided into features and labels before splitting into training and test sets using the 'train_test_split' method from 'sklearn'. 'stratify = y_cardio_df' is used to ensure that both training set and test set have equal rows of label data. The size of the test set is 20% of the entire dataset. Age data remain right-skewed so, discretising them is beneficial to ensure each age group gets equal representation.

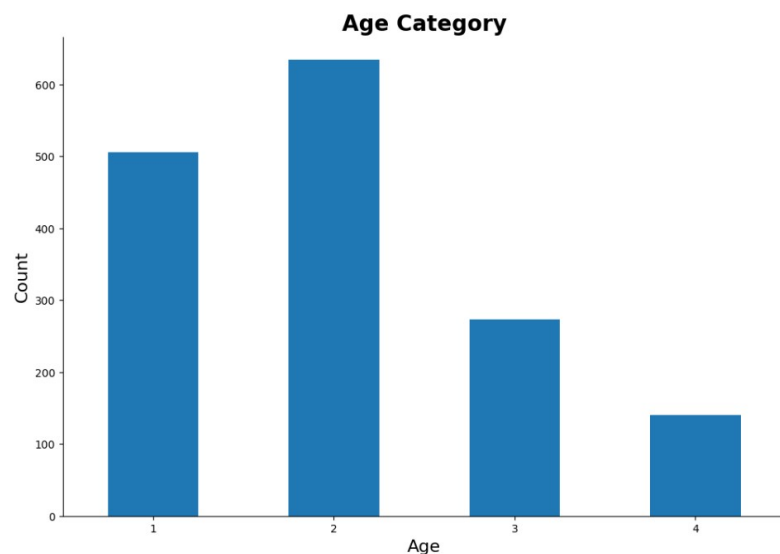


Figure 2.9: Age Category

In our chosen features, there are 6 features with nominal data and 3 features with ordinal data. In the 6 features with nominal data, 5 – gender, family, junk food, smoking, and discipline - will be preprocessed with “LabelBinarizer” because their data consists of yes or no, or male or female. “OneHotEncoder” would not be suitable as it just adds unnecessary complexities when dealing with binary features. Transportation will be encoded with “OneHotEncoder”.

“OrdinalEncoder” will be used on alcohol, tv, and snack. Finally, for better performance, cardiovascular risk will also be encoded with ordinal encoder.

2.4 Model Selection

Based on the dataset presented, this is a supervised, offline, multi-class classification problem. The models selected are, k-nearest neighbors, an extended logistic regression, and random forest.

2.4.1 K-Nearest Neighbors

Neighbours-based classification does not construct a model to make predictions but stores instances of the training data. In K-Nearest Neighbors, no models are generated; instead, learning is done instance-based. ScikitLearn offers two different neighbours-based classification options which are `KNeighborsClassifier` and `RadiusNeighborsClassifier`.

K-nn can also handle multiclass classification problems. Additionally, k-nn works well with a relatively small dataset of only 1.9 thousand rows of data. K-nn is a non-parametric algorithm, meaning it does not assume any underlying data distribution. This makes them more versatile for various data types and distributions.

K-nn makes predictions based on simple majority vote of the nearest neighbors of each point. The number of nearest neighbors to consider is important when classifying the output. A high number of neighbours to consider means the algorithm is more sensitive to the local structure, subtle patterns and variations in the data but, the is more likely to overfit while a lower number may give better generalization but risk underfitting. Additionally, to determine whether a point is to be regarded a neighbour, distance metrics are used. There are 3 common distance metrics to consider, Euclidean distance, Manhattan distance, and Minkowski distance.

2.4.2 Logistic Regression

Logistic regression is naturally a binary classifier, meaning it is designed to predict one of two possible outcomes. However, logistic regression can be extended to handle multiclass classification via several approach. The approach we used is One-vs-Rest or One-vs-All.

In OvR, 3 binary Logistic regression model is trained, one for each cardiovascular risk (low-risk detector, medium-risk detector, and high-risk detector). Then, to classify an individual, the class whose classifier outputs the highest score is selected.

2.4.3 Random Forest

Random Forest is a machine learning algorithm can be used for classification or regression task, creating a collection of decision trees (forest), each trained on random subsets of the data. It performs well on small datasets by averaging multiple decision trees, which reduces overfitting.

Moreover, its robustness to noisy data improves accuracy and stability, while its ability to manage multi-class classification makes it suitable for three-category task (low, medium, high). This combination of simplicity, interpretability, and performance makes Random Forest a strong option.

In addition, random forest is helpful for feature selection. It ranks the importance of each feature and help determine which features have the greatest impact on the prediction, making it useful in data analysis and interpretation.

2.5 Model Training and Validation

The main performance measure used to assess the models are confusion matrix, precision, recall, F₁ Score, and AUC-ROC graph.

K-fold cross validation is used when in training the model. This is done manually with StratifiedKFold to give us more control over the type of metrics to measure in assessing the performance of the model.

2.5.1 K-Nearest Neighbors

Training is done with the default hyperparameters to assess how well the basic model does on the training set. 5 splits are used for the k-fold cross validation. The default hyperparameters for sklearn's knn model are **5 numbers of neighbours**, **uniform** weight, and **Euclidean** distance metric.

Cross Validation metrics:

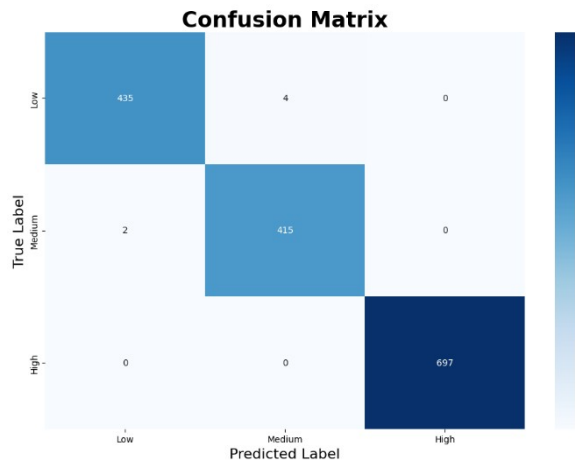
Metrics (weighted)	Average Values
Accuracy	0.987122
Precision	0.987474
Recall	0.987122
F ₁ Score	0.987110

A final model is trained once again on the entire training set.

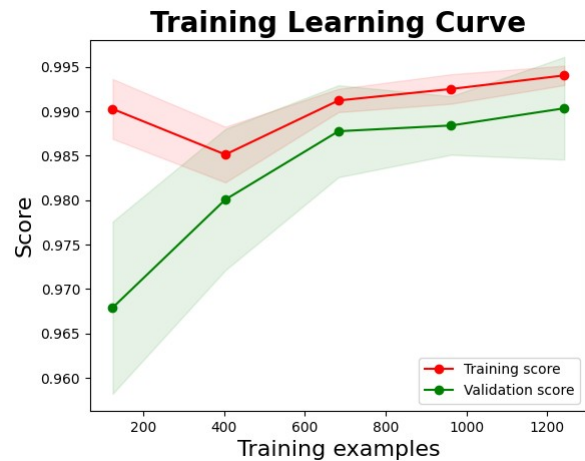
Metrics after training on the entire training set:

Metrics (weighted)	Values
Accuracy	0.996137
Precision	0.996143
Recall	0.996137
F ₁ Score	0.996137

The model maintains its performance even after training on the entire training set.



[Figure 2.10: KNN Training Confusion matrix](#)



[Figure 2.11: KNN Training Learning Curve](#)

According to the confusion matrix, most misclassification occurs in the low-risk – 4 false negatives and 2 false positives - and medium-risk – 2 false negatives and 4 false positives - category. The learning curve shows that the training score starts high and remains consistent throughout, except for a small initial dip, before stabilising at a high level. Such performance indicates some overfitting initially.

The validation score starts much lower than the training score but, as the number of samples increases, the validation score improves significantly, showing that the data generalises well as it sees more data. The gap between training score and validation score also decreases as the training examples increase, indicating improved generalization.

The shaded areas represent the uncertainty in the model's performance. The training performance is very stable as indicated by the narrow red shaded area but, the model's performance on the validation set fluctuates more when examples are small as denoted by the wider green shaded area. However, the uncertainty does reduce as more training data is added, indicating that the model is becoming more stable at generalising.

2.5.2 Logistic Regression

At first, the training set was preprocessed and divided into 5 folds with using Stratified K-Fold Cross-Validation. To ensure have a fair evaluation, the Logistic Regression model with the following parameters: **solver=lbfgs**, **max_iter=5000**, **penalty=l2** and **class_weight=balanced** was with 4 out of 5 folds, while the remaining folds acts as the validation. The results are showed below.

Cross Validation metrics:

Metrics	Average Values
Accuracy	0.984537
Precision	0.984590
Recall	0.984536
F ₁ Score	0.984496

After that, the final model was trained with the entire training set.

The results are displayed below. Model evaluation after training with the training set:

Metrics	Values
Accuracy	0.992268
Precision	0.992256
Recall	0.992268
F ₁ Score	0.992260

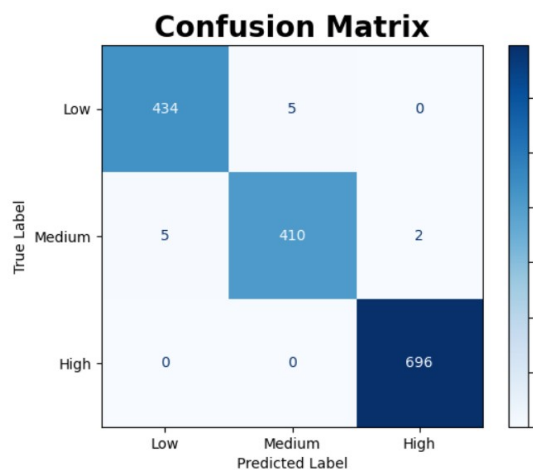


Figure 2.12: Logistic Regression Training Confusion Matrix

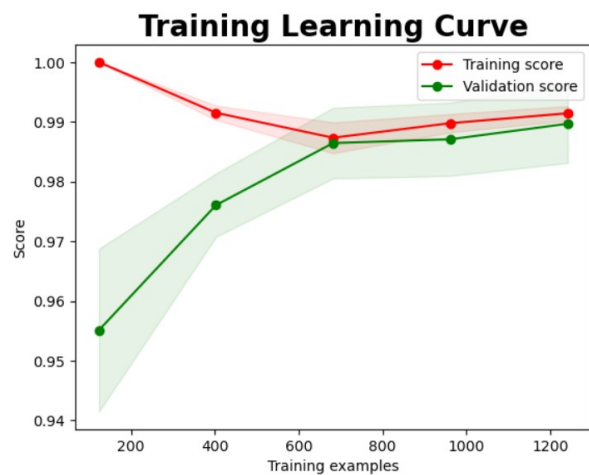


Figure 2.13: Logistic Regression Training Learning Curve

Looking at both results, the final model has a slightly better performance when training with the entire training set.

Based on the confusion matrix, the model makes the most misclassifications when attempting to predict medium-risk category with 7 false negatives and 5 true positives.

According to the learning curve, the model is initially overfitting, with a very high training score, but dips as the number of data introduced increases. This means that the model's ability to generalise is poor.

The validation score starts out much lower than the training score but improves as more data is used, indicating the model's improvement at generalisation. Finally, the gap between training score and validation score at the end does reduce, but it would still benefit from more data or fine-tuning to further close the gap.

The training performance is very stable as indicated by the narrow red shaded area but, the model's performance on the validation set fluctuates more when examples are small as denoted by the wider green shaded area. The variance fluctuates the most between 400 and 800 training samples. However, the uncertainty does reduce as more training data is added, indicating that the model is becoming more stable at generalizing.

2.5.3 Random Forest

To evaluate the performance of the Random Forest model on the training set, a 5-fold cross-validation is applied using the default hyperparameters. In sklearn's Random Forest classification model, the default settings include **100 decision trees**, **Gini** impurity as the criterion for splitting, and **no limit on tree depth**, allowing the trees to grow until all leaves are pure or contain fewer than the minimum number of samples required for splitting.

Cross Validation metrics:

Metrics (weighted)	Average Values
Accuracy	0.999517
Precision	0.999518
Recall	0.999517
F ₁ Score	0.999517

The final model is retrained using the complete training set.

Metrics after training on the entire training set:

Metrics (weighted)	Values
Accuracy	0.999356
Precision	0.999358
Recall	0.999356
F ₁ Score	0.999356

The model sustains its performance even after being trained on the full training set.

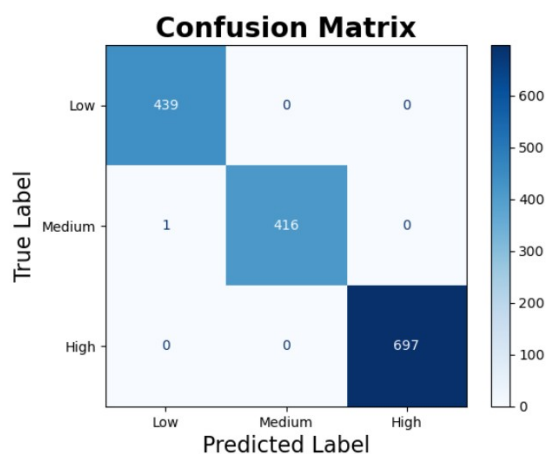


Figure 2.14: Random Forest Training Confusion matrix

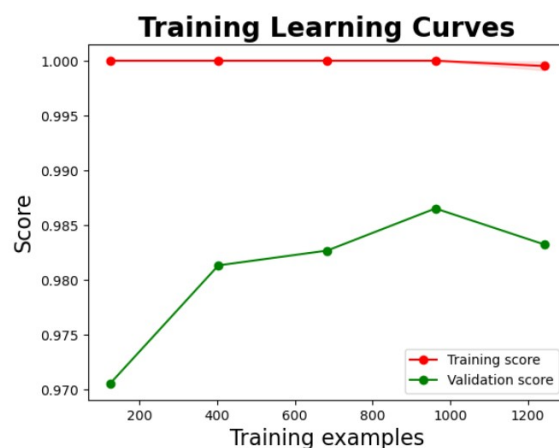


Figure 2.15: Random Forest Training Learning Curve

The confusion matrix in Figure 2.14 demonstrates that the model's predictions across three classes: Low, Medium, and High. The majority of the predictions are accurate, with 439 correctly classified as Low, 416 as Medium, and 697 as High. There are minimal

misclassifications, with only one instance of Medium being classified as Low. This indicates that the model has successfully learned to distinguish between these categories with high accuracy.

On the training learning curve in Figure 2.15, the training score remains near 1.0 across different sample sizes. The validation score increases as more examples are added but shows a slight gap from the training score, indicating some overfitting. However, the gap between the training and validation scores narrows as more data is added, indicating that the model's generalization improves with additional training examples, reducing uncertainty.

2.6 Model Tuning and Testing

To properly assess whether the model has overfitted, the model will be used on the test set. Before the test set is used, it will be preprocessed with the same encoders created on the training set to look like the training set to ensure uniformity and consistency.

2.6.1 K-Nearest Neighbours

After running the model on the test set, the model recorded consistent performance.

Testing Metrics:

Metrics (weighted)	Average Values
Accuracy	0.987147
Precision	0.987736
Recall	0.987147
F1 Score	0.987148

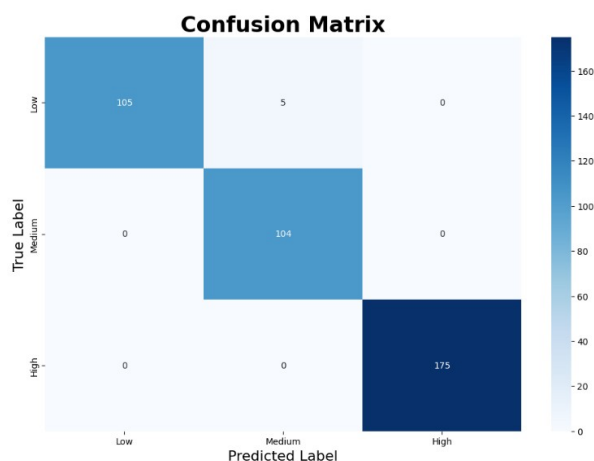


Figure 2.19: KNN Testing Confusion Matrix

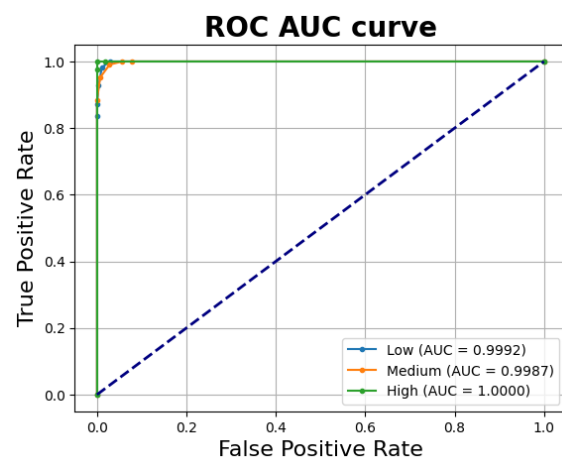


Figure 2.20: KNN ROC curve

Confusion matrix and learning curve are plotted again to better understand the model's performance. Overall, the model does well on both testing and training set with default parameters, indicating that the model has not overfitted.

However, misclassification still happens in the low-risk – 5 false negatives - and medium-risk – 5 false positives - categories. The ROC AUC curve shows good performance as well.

2 main hyperparameters – n_neighbours and distance metric - will be considered during fine-tuning. The fine tuning will be done with GridSearch. After fine tuning, it is found that the best KNN model uses **Manhattan distance metric** and considers **8 nearest neighbours**.

F₁ score improves slightly with the new, fine-tuned model but precision dips slightly.

Tuned Metrics:

Metrics (weighted)	Average Values
Accuracy	0.987147
Precision	0.987387
Recall	0.987147
F ₁ Score	0.987187

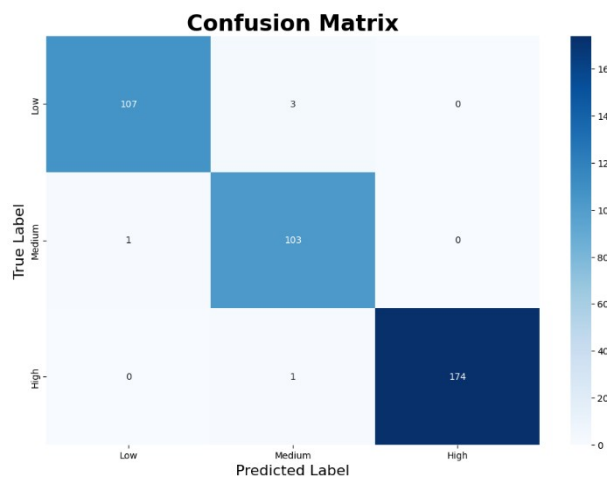


Figure 2.2418: KNN Fine-tuned Confusion Matrix

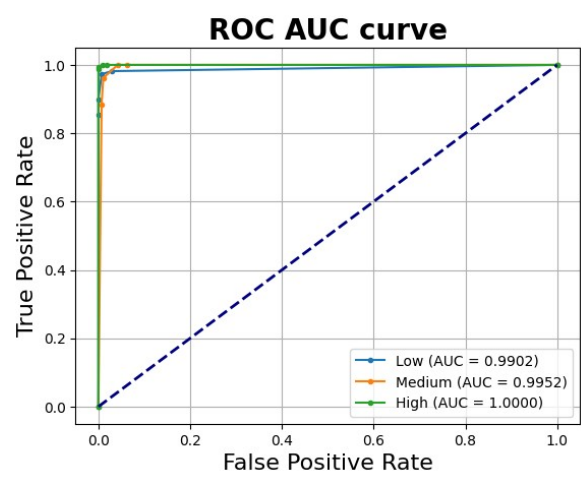


Figure 2.2219: Fine Tuned ROC AUC curve

According to the fine-tuned model, the most misclassifications still happen in medium-risk category and low-risk category with 1 false negative and 4 false positives. However, looking at the ROC AUC curve, the AUC for low-risk category dipped but improved slightly for medium risk category. Overall, the performance of the fine-tuned model improved with a rather negligible amount.

2.6.2 Logistic Regression

The model will be tested with the pre-processed test set. The results are showed below.

Testing Evaluation:

Metrics (weighted)	Values
Accuracy	0.989717
Precision	0.989708
Recall	0.989717
F1 Score	0.989703

By comparing with the Training Confusion Matrix, it is noticeable that the number of misclassifications for predicting the medium-risk category had decreased. This shows that the model performs better and generalize well to unseen data during the testing phrase. For the ROC AUC Curve, the model demonstrates a good performance in in distinguishing between the three categories.

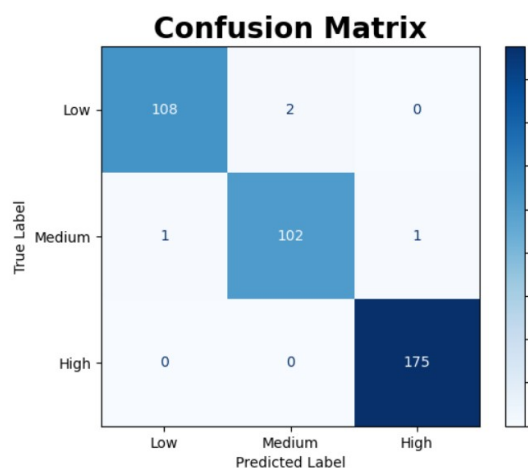


Figure 2.20: Logistic Regression Testing Confusion Matrix

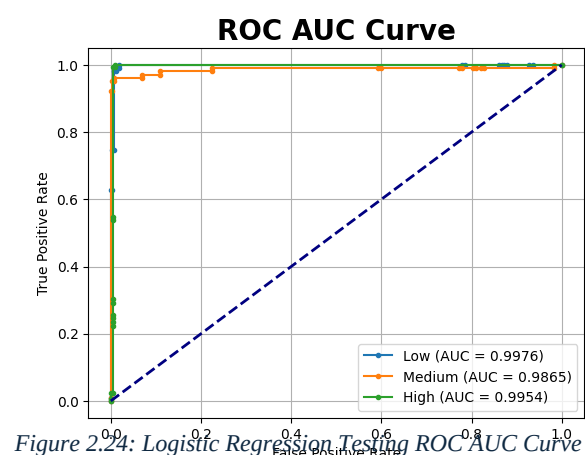


Figure 2.21: Logistic Regression Testing ROC AUC Curve

During fine tuning stage, we are focusing on five main hyperparameters: **regularization strength, penalty, maximum iterations, solver options and class weight.** Since the model is extended to One-vs-Rest (OvR), only the solvers that support this multiclass classification are used. Additionally, we have chosen not to use the 'sag' and 'saga' solvers because our dataset contains only 1.9k samples (after trimming), which considered a rather small dataset, and these solvers are better suited for larger datasets. Since the solvers used

support separate penalties, fine-tuning is done separately for both solvers. “liblinear” can support both l1 and l2 penalty while “lbfgs” can only support l2 penalty.

After performing fine tuning, it is shown that the best parameters to form the best Logistic Regression are **C: 0.1, max_iter: 5000, penalty: l2, solver: lbfgs, class_weight: balanced.**

lbfgs metrics:

Metrics (weighted)	Values
Accuracy	0.989717
Precision	0.989708
Recall	0.989717
F1 Score	0.989703

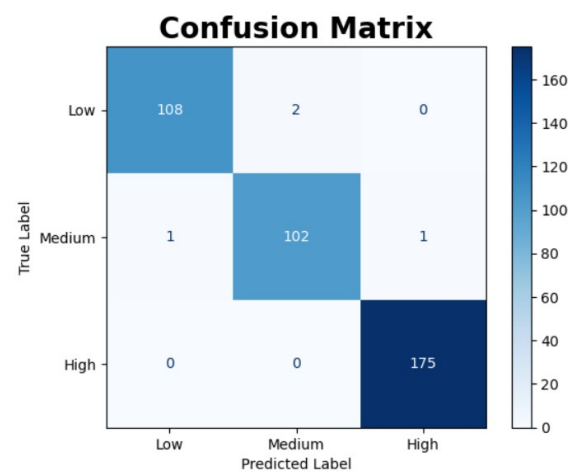


Figure 2.2522: Fined Tuned Confusion Matrix

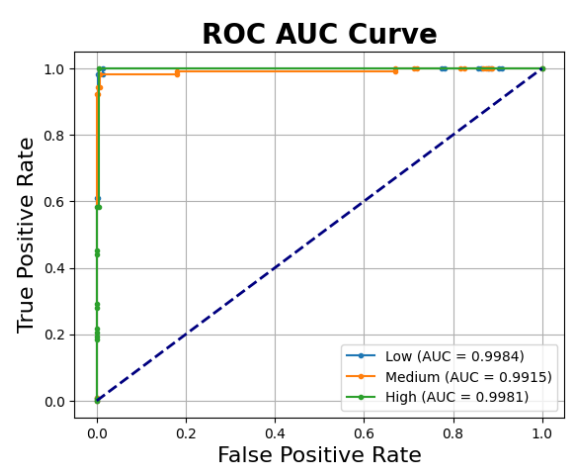


Figure 2.2623: Fined Tuned ROC AUC Curve

The Fined Tuned Confusion Matrix is similar to the Testing Confusion Matrix. Based on the ROC AUC Curve, there is a small improvement in the model's ability to differentiate these 3 categories. However, for the low-risk category continues to have the lowest score among the 3 categories.

The best parameter for the other solver, liblinear, are **C: 0.1, max_iter: 5000, penalty: l1, solver: liblinear, class_weight: balanced**. However, this version does not perform as well as the former.

liblinear metrics:

Metrics (weighted)	Values
Accuracy	0.958869
Precision	0.958757
Recall	0.958869
F ₁ Score	0.958804

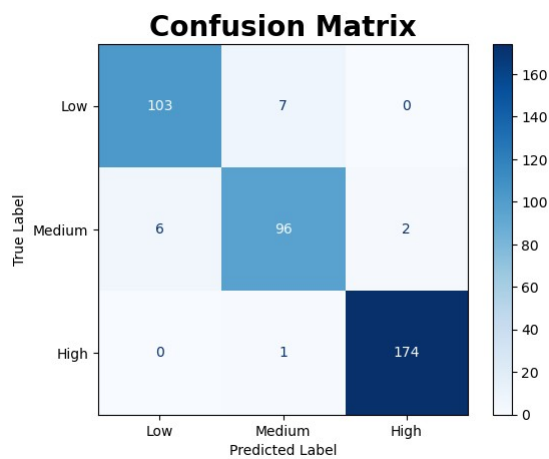


Figure 2.2724: Fined Tuned Confusion Matrix

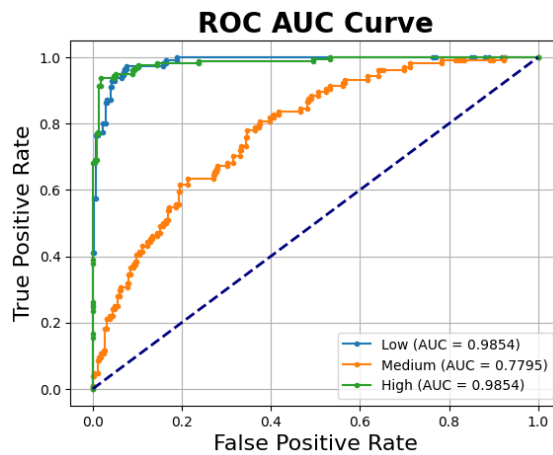


Figure 2.2825: Fined Tuned ROC AUC Curve

The model makes the most mistakes in distinguishing between low and medium categories. The ROC AUC curve shows that the model has a harder time predicting medium-risk category compared to predicting low-risk or high-risk categories. According to the confusion matrix, 7 false positives are made and 6 false negatives are made when classifying medium-risk category.

2.6.3 Random Forest

After running the model on the test set, the model recorded consistent performance.

Testing Metrics:

Metrics (weighted)	Average Values
Accuracy	0.992288
Precision	0.992313
Recall	0.992288
F ₁ Score	0.992289

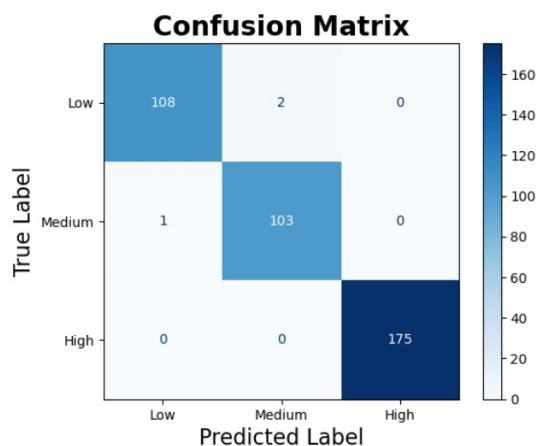


Figure 2.26: Random Forest Testing Confusion matrix

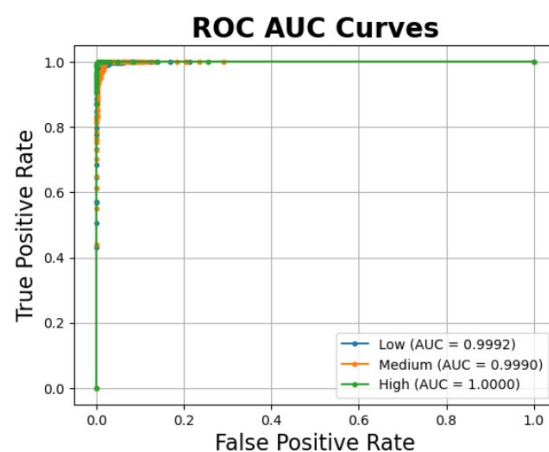


Figure 2.27: Random Forest ROC curve

Continuing with Figure 2.26, the confusion matrix for the test data shows that the model performs well but not perfectly when applied to new data. Specifically, 108 instances of Low risk were correctly classified, with 2 misclassified as Medium and none as High. For the Medium risk category, 103 instances were correctly predicted, while 1 Low was misclassified. In the High-risk category, the model correctly classified all 175 instances with no misclassifications. While the misclassifications in the Low and Medium categories are minimal, the overall performance remains very strong.

The ROC AUC curve in Figure 2.27 provides further insight into the model's performance, showcasing nearly perfect discrimination between the risk categories. The area under the curve (AUC) values for Low, Medium, and High are 0.9992, 0.9990, and 1.0000, respectively. These near-ideal AUC scores indicate that the model has excellent predictive

capabilities, with a very low false positive rate and a high true positive rate across all risk categories.

To achieve the best model performance, fine-tuning was conducted using grid search cross-validation over a range of hyperparameters. This process involved evaluating 576 different combinations of parameters, using 5-fold cross-validation, which resulted in a total of 2,880 model fits. The grid search explored parameters such as `max_depth`, `max_features`, `min_samples_leaf`, `min_samples_split`, and `n_estimators`, each impacting the complexity and performance of the Random Forest model.

Testing Metrics:

Metrics (weighted)	Average Values
Accuracy	0.992288
Precision	0.992313
Recall	0.992288
F ₁ Score	0.992289

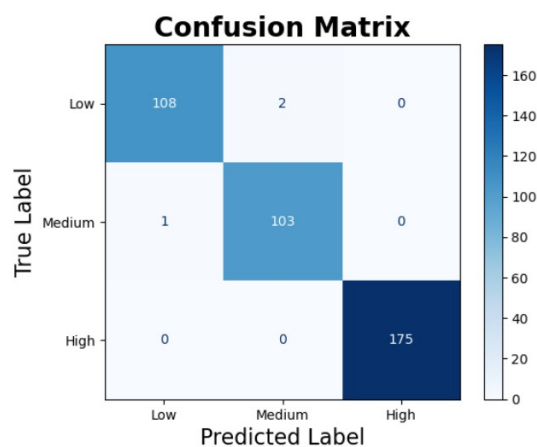


Figure 2.28: Random Forest Testing Confusion matrix

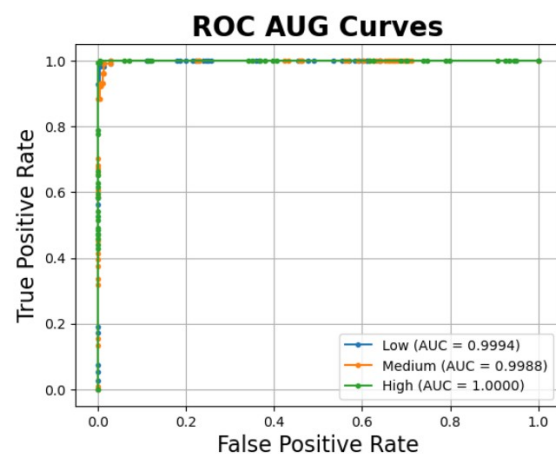


Figure 2.29: Random Forest ROC curve

The confusion matrix in Figure 2.28 remains the same as Testing, while the ROC AUC scores for tuned model presented in the Figure 2.29 also shows near-perfect performance. However, there is a slight variation in the AUC values between the two testing scenarios. The AUC for the Low category slightly increased from 0.9992 to 0.9994. Conversely, the AUC for the Medium category decreased from 0.9990 to 0.9988. The AUC

for the High category remains consistently perfect at 1.0 in both cases, indicating exceptional performance in distinguishing the High class from the others across both tests.

3.0 Results and Discussion

According to the fine-tuned version of each model,

Models	K-NN	Logistic Regression (solvers)		Random Forest
		liblinear with l1 penalty	lbfgs with l2 penalty	
Recall (weighted)	0.987147	0.958869	0.989717	0.992288
Precision (weighted)	0.987387	0.958757	0.989708	0.992313
F1 Score (weighted)	0.987187	0.958804	0.989703	0.992289

According to these metrics, the random forest model performs the best with the highest values across all metrics, followed by logistic regression model with “lbfgs” solver and k-nn comes in last. The logistic regression model with “liblinear” solver performs the worst amongst all 4, albeit recording values above 0.9.

According to the confusion matrices, random forest makes the least mistakes in predicting cardiovascular risks whereas logistic regression with “liblinear” solver makes the most mistakes. Nonetheless, all of these models have something in common, most misclassifications in prediction happens between low-risk and medium-risk categories compared to high-risk categories. This may be attributed to the uneven distribution – the domination of high-risk category - of the cardiovascular risks in the dataset, leading to poorer performance on the minority classes. This may also be the reason as to why “liblinear” solver performs so poorly; it is more sensitive to the distribution of classes.

3.1 Feature Importance

K-NN

Since K-NN is a distance-based model, it does not calculate feature importances.

Logistic Regression

In Logistic Regression, it does not have the “feature importance” metric like Random Forest. However, the feature coefficient of the model can provide the same insight as the “feature importance” metric to indicate the influence of that feature is.

Random Forest

Feature	Importance
BMI	0.722390
Snack	0.073774
Family_history	0.066922
Age_category	0.036632
Junk_food	0.017905
Exercise	0.014685
Alcohol	0.012645
TV	0.010880
Gender	0.010778
Transportation_bus	0.008732
Discipline	0.008573
Transportation_car	0.007792
Transportation_walk	0.004583
Smoking	0.003164
Transportation_motorcycle	0.000376
Transportation_bicycle	0.000170

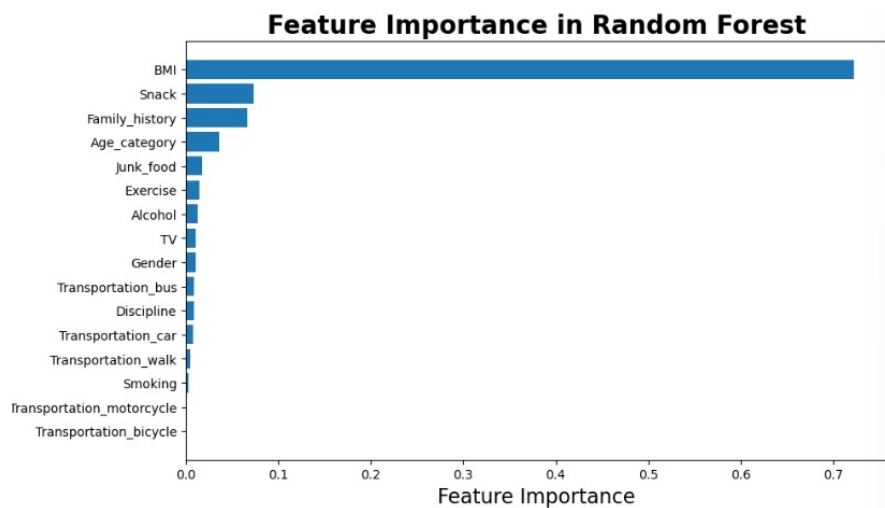


Figure 2.30: Random Forest Feature Importance

BMI is the most important factor in predicting cardiovascular risk, with an importance score close to 0.75, indicating it has a major influence on the model's performance. Following **BMI**, **Snack** and **Family History** show some importance, though their scores are much lower, both around 0.1, suggesting a moderate impact. **Age Category** also contribute slightly, while other lifestyle-related features such as **Junk Food** or **Transportation** play a moderate role but not as impactful as **BMI**, **Snack**, **Family History**, and **Age Category**.

3.2 Strengths and Weaknesses

KNN

Strengths

It is a simple model that is easy to implement and understand. No assumption is made about the distribution of the data, making it useful for data that does not follow standard distributions.

Weakness

K-nn can be affected by the dimensionality of data, particularly, datasets with a lot of features. As the number of features grow, the volume of the space grows, causing the available data points to be farther apart from each other. This makes it difficult to find meaningful nearest neighbours, as there may be fewer neighbouring points within a given distance. Furthermore, as the number of dimensions increases, the distance between any two points tends to become more uniform due to the diminishing influence of any single dimension.

Logistic regression

Strengths

Logistic regression is a powerful and easy implement model that uses probabilistic interpretation of classification. It works well with linearly separable data and can handle multiple class through the methods such as multinomial or one-versus-rest (OVA) regression. At the same time, it has L1 and L2 techniques to be regularized to address overfitting issue.

Weakness

However, logistic regression has a few main limitations. It assumes a linear relationship between the features and the outcomes. This means that it might not fit real- world data that comes with more complex pattern. Moreover, it is not suitable for predicting continuous values and classify non-linearly separable data.

Random Forest

Strengths

Random Forest classification model can be an effective choice for analyzing and predicting results especially for a dataset with less than 2.1k rows and 17 features. This method utilizes an ensemble of decision trees to make predictions based on various feature combinations, significantly reducing the risk of overfitting. One of the main strengths of it is ability to evaluate feature importance, which helps determine which features have most significant impact on predictions.

Weakness

Random Forest model's complexity can result in longer computation times, particularly as the size of the dataset or the number of trees increase. This may be a limitation when working with larger or more computationally demanding task.

4.0 Conclusions

In this analysis, we aimed to classify cardiovascular risk levels into low-risk, medium-risk, and high-risk using different classification models and methods such as k-nn, logistic regression, and random forest. Before training the model, we have selected relevant features that we feel have relationships with cardiovascular risk. Then, we have analysed the dataset to gain insights on the distribution of data and calculated descriptive statistics. Before removing outliers, we have combined height and weight into BMI. Finally, we have discretised age to ensure uniform representation of age.

From the three models we selected – k-nn, logistic regression, and random forest – after fine tuning, random forest performs the best with weighted recall of **0.992288**, weighted precision of **0.992313**, and weighted F score of **0.992289** after fine tuning. However, this model requires the most resource to run as, evidenced by its lengthy runtime.

Finally, in our logistic regression model, two solvers – “liblinear” and “lbfgs” – were used. Among these two, the model with “lbfgs” solver performs better, with **0.9897** in all three metrics whereas the model with “liblinear” solver does not perform as well, with **0.958869** weighted recall, **0.958757** weighted precision score, and **0.958804** weighted F_1 score, all of which are lower than the metrics in other models. Choosing the best solver, logistic regression model comes in second place in its predictive ability

In final place, k-nn records predictive ability with **0.987147** weighted recall, **0.987387** weighted precision, and **0.987187** weighted F_1 score after fine tuning. However, its performance is dependent on the choice of the number of neighbours to consider and the number of dimensions chosen to train it.

Regardless, each model performs well with every metric recorded exceeding 0.9. However, there is a caveat to consider, the analysis and performance of the model was heavily

limited by the dataset's size and the distribution of data. These models may not perform as well when tested on much larger, complete, and foreign datasets.